System Analysis and Design

Group Project — Assignment 1

Part I: Ideate & Requirements Analysis | Weight: 30% | 2026-2027, 1st Semester

For Part I of the project, you should provide a project idea submission (in any format that clearly conveys it), followed by a Software Requirements Specification (SRS) document, related use case and business process models, and supporting agile requirements artifacts such as personas, journey maps, or user stories.

This document covers Part I only. Parts II (Design) and III (Design Refinement & Implementation) will be issued separately, after Part I feedback has been incorporated.

1. Project Scope

Each team (no more than 4 students) will propose, specify, and later design a non-trivial software system. Choose any direction that gives you a real problem to reason about. The goal is good system-design practice, not a specific technology. Here are some candidate subjects for your reference:

- Productivity and collaboration: task management, scheduling, team workspaces, document collaboration
- E-commerce and marketplace: online retail, service booking, subscription management
- Health and wellness: appointment booking, fitness tracking, telemedicine, chronic-disease management
- Education and learning: adaptive quizzes, online tutoring, skills tracking, course management
- FinTech and business tools: personal finance, invoicing, expense reporting, budget planning
- Smart city / IoT services: transport information, environment monitoring, public-resource management
- Content and media platforms: blogging, podcast management, portfolio or creative-asset tools
- AI-assisted or agentic systems: an assistant or agent that reasons over an LLM and calls tools/APIs to complete tasks, e.g. a support agent, a research assistant, a scheduling agent

***Note**: AI or agentic components are welcome but not mandatory. Later parts (Design and Design Refinement & Implementation) will ask you to build the solution using agentic development regardless of what your system is. That's independent of whether your system itself is an agent.*

2. AI Tools Policy

Use of AI tools is encouraged. Every submitted report must include an AI Tools Reflection covering:

Tool chain	Tools used and the specific task each served
Cost	API usage, subscription fees, tokens, or free-tier limits reached; and human working hours
Pros / cons	Strengths and limitations of each tool for your tasks
Verification	How you validated AI-generated content

Enhancement	What you changed, corrected, or extended beyond the AI output
Artifacts	Prompts, skills, or any other scripts used

3. Structure and Process

First, your team submits a short idea write-up (only volunteer teams present their idea in class); then, after feedback, your team produces the full Software Requirements Specification (SRS).

3.1 Idea Submission (Ideate)

Submit to	Canvas (all teams, any format that clearly conveys your idea)
Deadline	Before the in-class idea session, **2026/10/12,13:30**
In-class presentation	Volunteer teams only, ~8 minutes + Q&A each (sign up in advance; slots are limited)

Any initiative or agile way to present the idea is encouraged: a short written brief, a slide deck, a lean/business-model canvas, a sketch or mockup, a short video, whatever gets the idea across most clearly. A full write-up is not required; cover the points below in whatever form suits your idea.

Idea submission — what to cover

- Identity: title, team name, members (no more than 4 students)
- Problem and motivation: background, main goals, scope
- Solution: core functionality, key characteristics, intended users and usability goals
- Market context: survey of similar products — utilities, limitations, your novelty
- Technology: platforms, languages, libraries, frameworks, third-party services
- Plan: team organization, methodology (e.g. Scrum), collaboration tools, preliminary schedule
- Challenges: anticipated technical and organizational risks
- Professional growth: how this project benefits you personally
- Optional: reference book or articles; if AI/agentic components are included, briefly note their role

In-class presentation (volunteer teams)

- ~8 minutes per team including Q&A — all members present
- Use slides, diagrams, or a prototype sketch — choose what communicates best
- Live feedback from the instructor and class should shape everyone's thinking, not just the presenting team's

Note: Teams that don't present will receive written feedback on their Canvas submission.

3.2 Requirements Specification

Deadline	**2026/10/23, 19:00**
Submit to	Canvas — specification document + modelling files

Produce a Software Requirements Specification (SRS) that incorporates instructor/class feedback and your team's own reflection since the idea submission.

No.	Section
1	Table of contents
2	Introduction (400–800 words) — refined from proposal
3	Strategic analysis — SWOT/TOWS; business goals and initiatives
4	Roadmap — agile or MVP roadmap with timeline and milestones
5	Use case and business process modelling
6	Glossary (min. 10 domain terms)
7	Supplementary specification (non-functional requirements)
8	UI mockups (min. 4 screens with brief descriptions)
9	References (min. 1 domain book and/or 2 articles, each annotated 30–60 words)
10	Team member contributions
11	AI tools reflection

Notes on Section 5 — Use case and business process modelling

- Use case diagrams for all major interactions
- Concise descriptions (2–4 lines) for all use cases; detailed specifications for at least 5
- Activity or BPMN diagrams for primary business processes
- If your system includes an agent or AI-assisted component, also show what it can act on: its inputs/tools and any point where it hands off to a human

  Choose whatever notation (UML, BPMN, or other) serves clarity. Precision and communication matter more than notation compliance. Optional agile artefacts: user personas, journey maps, product backlog, epics, user stories, enabler stories.

Notes on Section 7 — Supplementary specification

Cover the usual non-functional categories — usability, security, maintainability, performance, reliability. If your system includes an agent or AI component, also address:

- Accuracy/reliability — acceptable error rate and how you'd check it
- Latency and cost — acceptable response time and rough cost per use
- Autonomy — what it may do unsupervised vs. what needs confirmation
- Privacy — what data it may access or send to a third-party provider